

Dr. FAWAD ALI MANGI (PhD Approved Supervisor, MUET)

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Department of Computer Systems Engineering, Mehran University of Engineering & Technology (MUET)
Jamshoro, Sindh, Pakistan

PROFESSIONAL SUMMARY

Academic professional with over 11 years of experience in teaching, research, and academic administration in Computer Systems Engineering. Specialized in process mining, formal verification, and data-driven system analysis. Demonstrated expertise in curriculum development, postgraduate program coordination, and supervision of research at undergraduate, postgraduate, and doctoral levels. Actively engaged in quality assurance, academic governance, and institutional development.

ACADEMIC POSITIONS

Assistant Professor (Ad hoc), **Department of Computer Systems Engineering, MUET**
December 2024 - Present

Lecturer, **Department of Computer Systems Engineering, MUET**
April 2015 – December 2024

ACADEMIC AND PROFESSIONAL RESPONSIBILITIES

- Designed and architected software systems and distributed computing environments for Software Development and Artificial Intelligence Labs.
- Deliver undergraduate and postgraduate courses including Software Development, Data Structures and Algorithms, Database Systems, Software Engineering, Machine Learning, and Cloud Computing.
- Design course materials, assessments, and laboratory work aligned with Outcome-Based Education (OBE) and accreditation standards.
- Supervise undergraduate Final Year Projects (FYP), MS theses, and co-supervise PhD research in software systems, AI, and data analytics.
- Provide academic advising and student mentoring for BS, ME, MS, and PhD programs.
- Contribute to curriculum design and revision as an active member of the Curriculum Review Committee (CRC).
- Coordinate postgraduate programs, ensuring academic quality, compliance, and research progression.
- Organize and manage academic activities including project exhibitions, workshops, and seminars.
- Implement quality assurance practices as Departmental ISO/Quality Coordinator.
- Provide academic advising and mentoring to undergraduate and postgraduate students in the capacity of Academic Counsellor, supporting course planning, academic progression, and career guidance.

TECHNICAL EXPERIENCE

- Design, develop, and maintain software systems and distributed computing environments for academic and research applications.

- Develop and manage web-based systems including departmental platforms and Learning Management Systems (LMS).
- Design and administer database systems (SQL/NoSQL), including schema design, optimization, and performance tuning.
- Implement software development lifecycle (SDLC): requirements analysis, system design, development, testing, and deployment.
- Establish and manage Artificial Intelligence Lab supporting machine learning pipelines and data processing systems.
- Maintain and upgrade computing infrastructure across multiple laboratories and servers.
- Develop full-stack conference management systems including submission, review, and registration modules.
- Apply software testing, validation, and debugging techniques to ensure system reliability and performance.

RESEARCH EXPERIENCE

- Conducted PhD research in process mining, formal verification, and probabilistic model checking.
- Developed frameworks for verification and validation of software/process models under uncertainty.
- Applied Statistical Model Checking (SMC) techniques for system reliability analysis.
- Focus on system validation, performance analysis, and reliability of complex software systems.

PUBLICATIONS

- [B-1] Explainable AI (XAI) in Air Traffic Monitoring System** – Machine Learning Based Air Traffic Surveillance System Using Image Processing (Book Chapter), 2026
- [J-1] Advancing Verification of Process Mining Models with Quantitative Model Checking in Stochastic Environment** – ITM Web of Conferences Journal, 2024
- [J-2] Statistical Model Checking in Process Mining: A Comprehensive Approach to Analyse Stochastic Processes** – Future Internet Journal, 2023
- [C-1] Integrating Process Mining with Probabilistic Model Checking via Continuous Time Markov Chains** – IEEE CSCE, 2023
- [C-2] PM2PMC: Probabilistic Model Checking Approach in Process Mining** – IEEE GlobConET, 2023
- [J-3] Analysis of Swarm Intelligence Routing Protocols in Large-Scale Mobile Ad Hoc Networks (MANET'S)** – Journal of Theoretical and Applied Information Technology, 2021
- [Under Review] Towards Enhanced Verification in Process Mining: A CTMC-Based Probabilistic Model Checking Approach**

ADMINISTRATIVE AND LEADERSHIP ROLES

Postgraduate Program Coordinator

- Oversee ME, MS, and PhD programs, ensuring academic quality, research progress, and compliance with regulatory frameworks.

- Manage Learning Management System (LMS), including course deployment and academic content delivery.

Academic Counsellor (BS Computer Science Program)

- Provide structured academic guidance to undergraduate students on course selection, degree progression, and career pathways.
- Support student retention and performance through mentoring and continuous academic monitoring.

Member, Industrial Advisory Board (IAB)

- Fostering department–industry partnerships and consultancy.
- Contribute to strengthening academia–industry collaboration and aligning educational programs with industry needs.

Departmental ISO / Quality Coordinator

- Implement and monitor quality assurance processes in accordance with Higher Education Commission (HEC) and Pakistan Engineering Council (PEC) standards.
- Develop and document academic and administrative workflows to improve departmental efficiency.

Final Year Project (FYP) Coordinator

- Manage planning, execution, and evaluation of undergraduate projects.
- Ensure alignment of projects with academic standards and industry relevance.

Conference Leadership Roles

- Served as Conference Secretary (INCCST'25) and Finance Secretary & Media Team Lead (IICT'26).
- Coordinated conference operations, including participant management and digital platforms.

PROJECT SUPERVISION

- Supervised undergraduate Final Year Projects (FYP) involving intelligent systems, machine learning applications, data-driven solutions, and mobile/web-based systems.
- Supervised Master's (MS/ME) research theses, guiding students in problem formulation, system design, experimentation, and technical documentation.
- Co-supervising PhD research scholar, providing support in advanced research design, methodology development, system modelling, and publication of high-quality research.
- Mentored students in developing innovative and socially impactful solutions, including assistive technologies and intelligent applications.

EDUCATION

PhD - Computer Science

March 2024

[University of Wollongong, Australia](#)

Thesis: Enhancing Verification and Validation of Process Mining Models under Uncertainty by Quantitative Model Checking Techniques

- Designed and implemented software models for process mining and probabilistic model checking.
- Focused on system validation and verification, performance analysis, and ensuring reliability of software systems.

Masters of Engineering - Computer Information Engineering

December 2014

Mehran University of Engineering and Technology, Jamshoro, Pakistan

Bachelor of Engineering - Computer Systems Engineering

December 2011

Mehran University of Engineering and Technology, Jamshoro, Pakistan

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, Python, Scala, Rust, Go, TypeScript.
- **Web & Front-End Technologies:** HTML5, CSS3, JavaScript, React.js, Angular.
- **Back-End & Frameworks:** Node.js, Spring Boot, Django, Flask, Git, Docker, Kubernetes, LMS platforms, Web hosting systems.
- **Databases & Data Management:** SQL, NoSQL (MongoDB, Firebase), Relational Databases, Cloud Databases.
- **Cloud & Distributed Systems:** AWS, Azure, GCP, Cloud Deployment, Microservices, Serverless Architectures.
- **Core Competencies:**
 - Software Development Lifecycle (SDLC)
 - System Architecture & Design
 - Software Testing & Debugging
 - Database Design & Administration
 - Machine Learning Workflows
 - Data Processing & Analytics

HONORS, AWARDS, AND SCHOLARSHIPS

- Awarded the **HEC HRDI-UESTP PhD Scholarship** by the Higher Education Commission of Pakistan, for pursuing PhD studies.
- Awarded the **International Postgraduate Tuition Award (IPTA)** by the University of Wollongong, Australia, for pursuing PhD studies.

CERTIFICATION AND PROFESSIONAL DEVELOPMENT

Memberships

- Registered Engineer – Pakistan Engineering Council (PEC)
- National Scientists Directory, Pakistan Scientific & Technological Information Center (PASTIC)

Certification

- Master Trainer – Faculty Professional Development Program (FPDP), Higher Education Commission (HEC), Pakistan
- Master Trainer – Faculty Professional Development Program (FPDP), Mehran University of Engineering & Technology, Jamshoro

Workshop/Seminars Attended

- Workshop on GenAI in Action: Empowering Higher Education Faculty, by UK-Pakistan Generative AI Nexus for Higher Education (UPGAIN), (15-16 December 2025)
- Workshop on Capacity Building in Online Education, School of Mathematics & Computer Science, IBA Karachi (January, 2025)
- Curriculum based Orientation Program at Higher Education Commission of Pakistan (2018)
- Data Science at Department of Computer Systems Engineering, Mehran University of Engineering & Technology, Jamshoro (2018)
- Capacity Building Workshop for Outcome Based Education & Accreditation (one CPD credit point) organised by Pakistan Engineering Council, at Mehran University of Engineering & Technology, Jamshoro (2018)

REFERENCES

Available upon request